Semantic Networks – OSP-5 Notes

Definition

A **Semantic Network** is a **graph-based knowledge representation** technique used in Artificial Intelligence to **represent concepts (nodes)** and the **relationships (edges)** between them. It is a way to store and retrieve structured knowledge in a form that resembles human associative memory.

Structure

- **Nodes**: Represent **objects**, **concepts**, or **entities** (e.g., "Dog", "Animal", "Tail").
- **Links / Edges**: Represent **relationships** between nodes (e.g., *is-a*, *has-a*, *part-of*).

Example:

csharp
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```
Dog — is-a → Animal
Dog — has → Tail
```

Key Relationships

Relationship Type	Description	Example

is-a	Represents class hierarchy	Dog **is-a** Animal
has-a	Represents possession or part	Car **has-a** Engine
instance-of	Represents a specific object	Fido **instance-of** Dog

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Features

- Easy to **visualize** and **understand**.
- Supports **inheritance**: A node can inherit properties from parent nodes.
- Allows **association-based reasoning**.

Example Network

Knowledge:
"Birds are animals. A canary is a bird. Birds can fly. Penguins are birds but cannot fly."

Semantic Network:

csharp
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```
Canary — is-a → Bird — is-a → Animal
Bird — can → Fly
Penguin — is-a → Bird
Penguin — cannot → Fly
```

Advantages

- Mimics **human cognitive structure**.
- **Flexible** and **extendable**.
- Good for representing **hierarchical relationships**.

Limitations

- Can become **complex** and **confusing** in large networks.
- Does **not handle uncertainty** well.
- Lacks formal logical rules for deep inference (unlike predicate logic).

Applications

Domain	Use Case
Expert Systems	Representing medical or legal knowledge
Natural Language	Word meaning relationships, parsing
Knowledge Graphs	Web data, search engines

Summary Table

Feature	Description

Structure	Graph (nodes + edges)
Represents	Concepts and their relationships
Supports	Inheritance and associative reasoning
Used In	NLP, expert systems, semantic web